SVKM'S NMIMS MPSTME • DEPARTMENT OF COMPUTER ENGINEERING • DESIGN EXPERIENCE III (AY 2026–27)

AGROLOOP

Closing the Loop Between Urban Food Waste & Regenerative Soils

A Closed-Loop Circular Bio-Economy Platform for Mumbai MMR & Peri-Urban Agro-Belts

6,500 MT / Day **Mumbai Waste Crisis** 3,800 MT wet food scraps dumped daily at Deonar and Kanjurmarg landfills causing toxic methane fires & leachate.	**< 0.4% SOC** **Soil Carbon Depletion** Palghar & Shahapur farm soils stripped of biological carbon; smallholders burdened with ■32k/acre chemical input costs.	**56.5% Gross Margin** **Circular Economics** ■6.30/kg collection & processing cost vs ■14.50/kg blended sales realization; profitable by Month 14 of operations.

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01 • PROBLEM STATEMENT

The Broken Linear Economy: Urban Waste Overflow vs. Rural Soil Starvation

Urban Food Waste Crisis
6,500 MT Daily Generation

• Mumbai generates 6,500–7,200 MT MSW daily; 3,800 MT is wet organic scraps.
• Deonar (120 ha) & Kanjurmarg landfills face catastrophic methane buildup.
• Rotting scraps cause spontaneous toxic fires & carcinogen emissions.
• BMC actively levying SWM 2016 penalties (■5k–■25k) on non-compliant bulk generators.

Rural Soil Carbon Collapse
< 0.35% Soil Organic Carbon

• Palghar & Western Maharashtra soils depleted of organic carbon (<0.4% vs 1.0% ideal).
• Severe soil compaction, poor water retention & secondary salinization.
• Smallholders spend ■25k–■38k/acre on synthetic fertilizers.
• Diminishing crop yields drive smallholder debt & rural distress.

The Missing Systemic Bridge
Failed Municipal Status Quo

• Zero verified logistics connecting city organic waste with rural farms.
• Central municipal compost has 40–50% plastic contamination; farmers reject it.
• On-site apartment composter machines have an 85% abandonment rate.
• Complete lack of data-driven agronomic soil advisory tailored to local crops.

02 • TARGET USERS & EXISTING GAP

Who Experiences the Problem & Why Do Current Solutions Fail?

Bulk Waste Generators (BWGs)
1,200+ Commercial Kitchens

• Hotels, cloud kitchens & corporate canteens in Bandra/BKC/Andheri.
• Pain: Strict BMC SWM 2016 mandates; threat of health license cancellation.
• Need: Reliable daily off-site pickup & audit-ready compliance proof.

Gated Residential Societies
400+ Large CHS (>200 Flats)

• High-density complexes in Powai, Thane & Goregaon.
• Pain: On-site composters fail due to stench, flies, and resident revolt.
• Need: Clean odor-free collection with monthly ESG diversion certificates.

Smallholder Farmers
14,000+ Peri-Urban Growers

• Vegetable, fruit & paddy farmers in Palghar & Shahapur clusters.
• Pain: Crushing chemical fertilizer expenditure and degraded hard soil.
• Need: Low-cost, pure organic bio-inputs with certified NPK and AI guidance.

THE CRITICAL INDUSTRY GAP: WHY STATUS QUO COLLAPSES
1. **Contamination:** Municipal composting yields 40%+ plastics; AgroLoop uses BLE IoT doorstep QC (<5% rejection).
2. **Odor Failure:** Aerobic units rot; AgroLoop uses Bokashi EM1 anaerobic fermentation dropping pH to 4.0 in 48h.
3. **Zero Guidance:** AgroLoop integrates Gemini 1.5 Pro AI with ICAR deterministic guardrails for precision crop dosage.

03 • PROPOSED SOLUTION

AgroLoop: The Closed-Loop Hyper-Local Circular Bio-Economy Platform

01. COLLECT Smart EV Logistics	02. PROCESS Mahape Bio-Refinery	03. DISTRIBUTE Direct Farm Market	04. ADVISE Gemini AI Advisor
• Daily scheduled pickups via 3-wheeler EV loaders. • Bluetooth digital load cell doorstep weight capture. • Instant optical purity grading (<5% tolerance threshold). • Gamified green points for clean households.	• Bokashi EM1 anaerobic fermentation (odor lock at pH 4.0). • Eisenia fetida vermicomposting (45-day cycle). • 550°C Biochar pyrolysis retort (100+ yr carbon sink). • Concentrated liquid bio-fertilizer extraction.	• Direct-to-farm marketplace bypassing middlemen. • Palghar & Shahapur FPO bulk delivery clusters. • 35% lower cost than synthetic DAP/NPK fertilizers. • Standardized 25kg/50kg FCO 1985 certified sacks.	• Google Gemini 1.5 Pro multimodal reasoning engine. • Deterministic ICAR agronomic safety rule engine. • Vernacular conversational advisory (Marathi/Hindi). • Custom soil pH balancing & crop stage dosage.

04 • TECHNOLOGY & IMPLEMENTATION

Production Full-Stack Architecture, Guarded AI & IoT Telematics

Full-Stack Architecture
Next.js 15 + TypeScript

• Next.js 15 App Router core with 60 generated pages.
• Supabase PostgreSQL backend with strict Row Level Security (RLS).
• Offline local storage fallback engine ensuring 100% reliable viva demo.
• Tailwind CSS cleantech design system & Recharts dynamic visualizations.
• Installable Progressive Web App (PWA) with service worker caching.

Guarded AI Agronomist
Gemini 1.5 + ICAR Guardrails

• Google Gemini 1.5 Pro integration for vernacular dialogues.
• Deterministic agronomic rule engine pre- and post-processor.
• Enforces ICAR package of practices dosage safety ceilings.
• Eliminates AI hallucinations (guarantees zero crop chemical toxicity).
• Converts farmer soil test deficits into exact vermi & biochar schedules.

IoT Scale & Telematics
BLE IoT + MapLibre GL

• Bluetooth Low Energy (BLE) load cell integration for doorstep weigh-in.
• MapLibre GL interactive geospatial tracking of Mumbai, Mahape & Palghar.
• Automated contamination slider penalizing dirty bins (>5% inorganic).
• Batch maturation state machine tracking fermentation days to harvest.
• Automated generation of MCGM-compliant digital SWM certificates.

05 • INNOVATION & MEASURABLE IMPACT

Robust Unit Economics, Carbon Sequestration & Farmer Prosperity

Unit Economics: 56.5% Margin
Dual-Revenue Engine

- EV Doorstep Collection Cost: ■2.80 / kg
- Bio-Processing & Curing Cost: ■3.50 / kg
- Total Unit Operating Cost: ■6.30 / kg
- Blended Selling Price: ■14.50 / kg
- **Gross Margin: 56.5%** (Profitable by Month 14)
- Dual cashflow: B2B subscriptions + bio-product sales.

Environmental Impact
1,875 MT CO■e Avoided / Hub

- 1,500 MT waste diverted from Deonar landfill annually.
- Eliminates 1,875 MT CO■e methane gas emissions.
- 550°C Biochar locks recalcitrant carbon for 100+ years.
- Verifiable voluntary carbon credits (VERs @ ■850/MT CO■e).
- Zero toxic leachate drainage into Thane Creek.

Farmer & Social Value
35% Farm Cost Reduction

- Replaces costly DAP & Urea (saves ■14,000/acre/year).
- Enhances soil water retention capacity by 28%.
- Proven 18% average crop yield increase across trials.
- Direct vernacular AI crop health advisory on WhatsApp.
- 25 dignified formal green logistics jobs per hub.

BUSINESS MODEL CANVAS

The Nine Building Blocks & Validated Willingness-to-Pay

Key Partners • MCGM & TMC Ward Offices • Gated Societies (Powai, Andheri) • Palghar Farmer Producer Orgs • EV Fleet Providers & KVK Palghar	**Key Activities** • Scheduled EV scrap harvesting • Doorstep optical purity grading • 4-Stream biological bioconversion • Gemini AI agronomist advisory	**Value Propositions** • **B2B:** 100% SWM compliance, zero stench, digital ESG certificates. • **Farmers:** 35% cheaper inputs, +28% moisture hold, +18% yield.
Customer Relationships • B2B Dedicated Account Managers • Society Green Soil Points Wallet • Vernacular WhatsApp crop alerts	**Channels** • Next.js 15 PWA Platform • 3-Wheeler EV Fleet Touchpoints • FPO Village Hubs & Retail Sacks	**Customer Segments** • 1,200+ Commercial Kitchens • 400+ Gated Societies (>200 flats) • 14,000+ Peri-Urban Smallholders
Cost Structure • EV Collection: ■2.80/kg | Bio-Processing: ■3.50/kg • FPO Cluster Transport: ■1.20/kg | Tech & AI API: ■0.30/kg	**Revenue Streams** • B2B Subscriptions: ■2.5k–■5k/mo per kitchen • Bio-Inputs (Blended ■14.50/kg) | Carbon Credits (■850/tCO■e)	**Financial Health** • Gross Margin: 56.5% | Breakeven: Month 14 • Yr 1: ■1.02 Cr | Yr 2: ■2.68 Cr | Yr 3: ■6.85 Cr

INCUBATION & EXTERNAL VALIDATION

INNOVATE 360° 3rd Position Award & Institutional Incubation Alignment

■ INNOVATE 360° (2026–27) AWARD OF ACHIEVEMENT

Secured 3rd Position Across MPSTME & SIG AI Pitch Challenge

- **Organized by:** Dept. of Computer Engineering & SIG AI MPSTME (4th Sept 2026).
- **Signatories:** Dr. Priti Rumao (HoD, Computer Engineering) & Prof. in-Charge.
- **Jury Score: 93.0 / 100** across 6 venture evaluation dimensions.
- **Jury Commendation:** *'Outstanding unit economics (56% margin), production Next.js 15 demo, and safe deterministic ICAR agronomy guardrails.'*
- Verified Certificate preserved as primary evidence for Review III.

TBI & INCUBATION PIPELINE

Pre-Incubation, Grants & Field Trials

- **AIC-NMIMS Pre-Incubation:** Formal application filed for Cohort 8.
- **KVK Palghar Partnership:** On-ground farmer field trials & NPK testing.
- **Startup India Seed Fund Scheme (SISFS):** ■20 Lakhs grant pipeline.
- **RKVY-RAFTAAR (Agri-Ministry):** ■25 Lakhs grant for Mahape pyrolysis kiln.
- **IP & Trade Secrets:** Proprietary ICAR inference algorithm protected.
- **Field Validation:** 12 smallholder trial plots verified in Manor/Palghar.

06 • FUTURE SCOPE & COMMERCIALIZATION

A Scalable 3-Phase Roadmap to Regional Multi-Hub Expansion

Phase 1 (Months 1–12)
Mahape Pilot Hub (MMR)

- Commission 5,000 sq ft Mahape hub.
- Deploy 4 EV loaders covering Bandra & Powai.
- Onboard 60 kitchens & 20 gated societies.
- Serve 650 smallholders in Palghar/Shahapur.
- Divert 900 MT waste / yr; ■1.02 Cr revenue.

Phase 2 (Months 13–24)
Commercial Breakeven (2 Hubs)

- Expand across Thane & Navi Mumbai belt.
- Corporate canteens (Reliance, Tata, Mindspace).
- Operational breakeven at Month 14.
- Deliver positive EBITDA (+■73 Lakhs) in Yr 2.
- Scale to 2,160 MT diversion & 1,800 farmers.

Phase 3 (Months 25–36)
Regional Scale (■6.85 Cr ARR)

- Containerized modular hubs in Nashik & Pune.
- Tokenized voluntary carbon credit sales.
- Scale to 5 regional hubs (5,400 MT/year).
- Achieve ■6.85 Cr revenue & +■2.25 Cr EBITDA.
- Replicate model to Smart Cities across India.

RAPID Q&A; DEFENSE PANEL

Pre-Loaded Answers to the Top 8 Evaluator & Judge Challenges

1. Contamination Control: Doorstep visual grading rejects bins with >5% inorganics. Households penalized green points. Trommel sieves at hub.	**2. Logistics Viability:** ■2.80/kg via 3-wheeler EVs (charging @ ■0.40/km vs ■3.80/km diesel). Route density keeps shifts <12 km.	**3. Farmer Adoption:** Farmers spend ■28–■35/kg for chemicals. AgroLoop vermi @ ■10 & biochar @ ■24 cuts input costs by 35% with +28% water retention.	**4. Odor & Leachate:** Bokashi EM1 anaerobic fermentation drops pH to 4.0 within 48h, completely halting foul ammonia and putrefactive odors.
5. AI Advisory Safety: Deterministic agronomy rule engine clamps Gemini outputs to ICAR package of practices, preventing toxic over-fertilization.	**6. Breakeven Timeline:** Operational breakeven reached at Month 14. Year 2 delivers +■73 Lakhs EBITDA supported by upfront B2B subscription retainers.	**7. SWM 2016 Alignment:** 100% compliant with SWM Rules 2016 Rule 4. Issues audited digital diversion certificates required by BMC health inspectors.	**8. Carbon Sequestration:** 550°C Biochar creates recalcitrant carbon sink (>100 yr half-life), generating verified carbon credits valued at ■850/MT CO■e.